

Process Flow: Bulk Export & Audit

Phase 1: The Generator (Auto_export_xl.js)

- **Initialization:** Load system settings, directory paths, and batch limits from the configuration file.
- **File Queuing:** Scan the `input_excel` folder and sort all valid `.xls/.xlsx` files alphabetically for processing.
- **Session Start:** Execute the Playwright `login()` function and store session cookies to avoid repeated logins.
- **Data Conversion:** Trigger `convertSingleExcelToJson` to transform the raw Excel file into a readable JSON format.
- **Row Validation:** Verify that every row contains an `Exporter Name` and skip any incomplete entries.
- **Portal Navigation:** Direct the browser to the `/export-ccm` URI while performing periodic session health checks.
- **Job Entry:** Click the "Add Job" button and input the Exporter name using an automatic retry/shorten logic.
- **File Upload:** Use the "Browse Files" component to upload the target internal Excel document to the portal.
- **Popup Mitigation:** Intercept and dismiss site-specific banners such as "Exchange Rate" or "Duplicate Invoice" warnings.
- **Form Progression:** Navigate through five strict form pages by monitoring native loading spinners rather than fixed timers.
- **Flat File Creation:** Click "Create FlatFile" on the final Review page and accept any final "Rewards" popups.
- **Secure Download:** Capture the server's `.sb` stream and save it to `output_sb` with a distinguishing `x` prefix.
- **Logging:** Write missing field data to `missing_fields_log.json` and system errors to `error_log.txt`.

Phase 2: The Validator (SB_compare.js)

- **Folder Targeting:** Define `input_sb` as the baseline directory and `output_sb` as the generated results directory.
- **File Pairing:** Match the baseline `abc.sb` file with the generated `xabc.sb` file using a prefix-matching algorithm.
- **Schema Loading:** Parse the Specs Excel file to establish table segments, mandatory flags, and data type rules.

- **Content Segmentation:** Divide the **.sb** text into internal table segments using **<TABLE>** markers and Record Separators.
- **Boundary Cleaning:** Terminate the parsing process exactly at the **<END-SB>** tag to exclude any trailing junk data.
- **Dynamic Filtering:** Exclude variable fields like **JOB NO.** and **DATE** that naturally differ between file versions.
- **Data Type Audit:** Validate that generated fields adhere to strict Numeric (**N**) or String rules defined in the Specs.
- **Length Verification:** Ensure no generated string exceeds the maximum character length permitted by the schema.
- **Parity Check:** Execute a 1-to-1 exact string equality comparison for all non-excluded values across both files.
- **Telemetry Generation:** Compile a comprehensive results array that prioritizes failed documents for immediate review.
- **Dashboard Rendering:** Pass the precise error coordinates to the Next.js UI to display interactive red/green status chips.

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